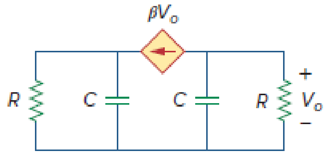
Problem

For what value of β is the circuit in Fig. 16.29 stable?

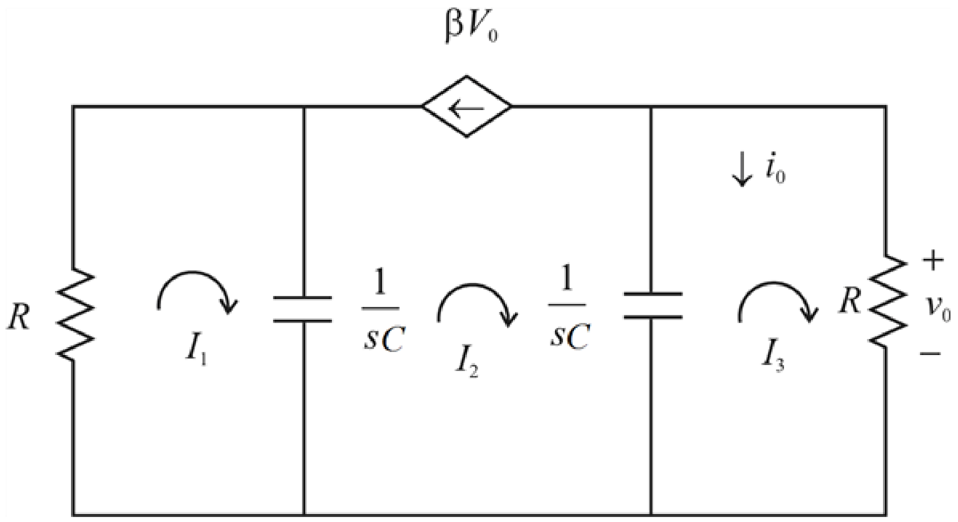
Figure 16.29:



Step-by-step solution

Step 1 of 6

The circuit in *s*-domain



Step 2 of 6

Applying mesh analysis to the first-order circuit at loop (1)

$$I_1 R + \frac{1}{sC}(I_1 - I_2) = 0$$

$$I_1\left(R + \frac{1}{sC}\right) - \frac{1}{sC} = 0 \quad ---(1)$$

At loop (2)

We know that

$$V_0 = I_3 R$$

$$I_2 = -\beta V_0$$

$$I_2 = -\beta R I_3$$

$$I_2 + \beta R I_3 = 0 \quad ---(2)$$

At loop (3)

$$\frac{1}{sC}(I_3 - I_2) + R I_3 = 0$$

$$I_3\left(R + \frac{1}{sC}\right) - \frac{1}{sC} I_2 = 0 \qquad ---(3)$$

Step 3 of 6

We can write equation (1), (2) and (3) in matrix form as

$$\begin{bmatrix} R+\frac{1}{sC} & \frac{-1}{sC} & 0 \\ 0 & 1 & \beta R \\ 0 & \frac{-1}{sC} & R+\frac{1}{sC} \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The determinant is

$$\Delta=\left(R+\frac{1}{sC}\right)\left\{R+\frac{1}{sC}+\frac{\beta R}{sC}\right\}+\frac{1}{sC}\{0-0\}+0$$

$$\Delta=\left(R+\frac{1}{sC}\right)^2+\frac{\beta \mathrm{R}^2}{sC}+\frac{\beta R}{s^2C^2}$$

$$\Delta=R^2+\frac{1}{s^2C^2}+\frac{2R}{sC}+\frac{\beta R^2}{sC}+\frac{\beta R}{s^2C^2}$$

$$\Delta=\frac{R^2s^2C^2+2RsC+\beta R^2sC+\beta R+1}{s^2C^2}$$

$$\Delta=\frac{s^2\left(R^2C^2\right)+s\left(2RC+\beta R^2C\right)+\beta R+1}{s^2C^2}$$

Step 4 of 6

The characteristics equation $(\Delta=0)$ gives the poles as

$$s^2\left(R^2C^2\right)+s\left(2RC+\beta R^2C\right)+\beta R+1=0$$

$$P_{1,2}=\frac{-\left(2RC+\beta R^2C\right)\pm\sqrt{\left(2RC+\beta R^2C\right)^2-4(\beta R+1)\left(R^2C^2\right)}}{2R^2C^2}$$

$$P_{1,2}=\frac{-\left(2RC+\beta R^2C\right)\pm\sqrt{4R^2C^2+4\beta R^3C^2+\beta^2R^4C^2-4\beta R^3C^2-4R^2C^2}}{2R^2C^2}$$

$$P_{1,2}=\frac{-\left(2RC+\beta R^2C\right)\pm\beta R^2C}{2R^2C^2}$$

Step 5 of 6

$$P_1 = \frac{-2RC - \beta R^2 C - \beta R^2 C}{2R^2C^2}$$

$$P_2 = \frac{-2RC - \beta R^2 C + \beta R^2 C}{2R^2C^2}$$

$$P_1 = \frac{-2RC - 2\beta R^2 C}{2R^2C^2}$$

$$P_2 = \frac{-1}{RC}$$

$$P_1 = \frac{-2RC(1+\beta R)}{2R^2C^2}$$

For circuit to be stable roots must be located left half of the *s* plane

$$\therefore \frac{-2RC(1+\beta R)}{2R^2C^2} < 0$$

$$1 + \beta R > 0$$

$$\beta > -\frac{1}{R}$$

Step 6 of 6

Which is negative when $\beta > \frac{-1}{R}$